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COMPUTER

CHAPTER 16

Internet

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CHAPTER 16

Internet

Computer Awareness

IN THIS CHAPTER

- Internet development
- IP addressing
- DNS and URLs
- Internet protocols
- Web browsers
- Web pages and domains

The Internet is a global network of interconnected computer networks that allows millions of computers and devices worldwide to communicate and share information.

It connects computers using communication technologies such as:

- Telephone lines
- Optical Fiber Cables
- Satellite Links
- Wireless Networks

The Internet enables sharing of:

- Data
- Information
- News
- Audio
- Video
- E-mails
- Online Services

Important Exam Point

Internet = Network of Networks

HISTORY OF INTERNET

ARPANET (Advanced Research Projects Agency Network)

- Developed in the USA.
- Started in 1969.
- Considered the foundation of the modern Internet.

Exam Point

- ARPANET is known as the predecessor of the Internet.

INTERNET IN INDIA

Public commercial Internet services in India were launched by VSNL on 15 August 1995; this was not the first use of computer networking in India.

- Videsh Sanchar Nigam Limited (VSNL)
- Internet Service Providers (ISPs)

Other Important Facts

- First political party website in India: Bharatiya Janata Party (BJP)
- First online telephone directory: Sikkim
- First high-speed rural broadband network: Idukki, Kerala

Exam Point

- Internet services started in India on 15 August 1995.

INTERNET RELATED TERMS

TCP/IP (TRANSMISSION CONTROL PROTOCOL / INTERNET PROTOCOL)

TCP/IP is the standard communication protocol used by the Internet.

It governs how data is sent and received between devices.

Why Important?

Without TCP/IP, Internet communication is impossible.

Exam Point

- TCP/IP is called the Backbone of the Internet.

HOST

A device connected to the Internet is called a Host.

Examples:

- Computer
- Laptop
- Smartphone
- Server

INTERNET SERVICE PROVIDER (ISP)

An ISP is a company that provides Internet access to users.

Examples

- Airtel
- Reliance Jio
- BSNL

Exam Point

- ISP = Internet Service Provider

IP ADDRESS(INTERNET PROTOCOL ADDRESS)

An IP address is a numerical address assigned to a network interface. Its uniqueness depends on the network scope; private addresses can be reused on separate networks.

It identifies the device's location on the Internet.

Example

192.168.1.1

109.12.31.115

Characteristics

- Must be unique within the relevant network scope
- Used for identification
- Assigned by ISP

Real-Life Example

Just as Aadhaar Number identifies a person,

IP Address identifies a computer.

TYPES OF IP ADDRESS

1. Static IP Address

- Fixed address
- Does not change

Used In

- Servers
- Business networks

2. Dynamic IP Address

- Changes periodically
- Assigned automatically

Used In

- Home Internet connections

3. Private IP Address

Used inside local networks.

Example:

192.168.1.1

4. Public IP Address

Visible on the Internet.

Assigned by ISP.

IP VERSIONS

IPv4

- 32-bit address
- Example:

192.168.1.1

Total Addresses

Approximately 4.3 billion IPv6

- 128-bit address
- Developed due to shortage of IPv4 addresses.

Example:

2001:0db8:85a3::8a2e:0370:7334

Exam Point

- IPv4 = 32-bit
- IPv6 = 128-bit

DNS (DOMAIN NAME SYSTEM)

DNS converts domain names into IP addresses.

Example

facebook.com

157.x.x.x

DNS helps humans remember website names instead of numerical IP addresses.

Exam Point

- DNS translates domain names into IP addresses.

URL(UNIFORM RESOURCE LOCATOR)

A URL is the address of a resource on the Internet.

Tim Berners-Lee developed the Web addressing approach around 1990; URL syntax was standardized in RFC 1738 in 1994.

URL Structure:protocol://hostname/filename

Example:https://www.facebook.com/Gurpreet

Components of URL

1. Protocol

Example:

- https
- ftp
- http

2. Host Name

Example: facebook.com

3. File Name / Path

Example: /Gurpreet

Exam Point

- URL = Address of a web resource.

WORLD WIDE WEB (WWW)

Also called:

- WWW
- W3
- Web

The WWW is a collection of interconnected web pages available on the Internet.

Important Difference Internet ≠ WWW

Internet = Infrastructure

WWW = Collection of web pages

COMPONENTS OF WWW

1. HTTP (HyperText Transfer Protocol)

Used for communication between web browser and web server.

2. URL

Address of web page.

3. HTML(HyperText Markup Language)

Language used to create web pages.

WEB SERVER

A Web Server is a powerful computer or software that stores website files and delivers them to users upon request.

Example

When you open Facebook:

1.Browser sends request.

2.Web server receives request.

3.Web server sends webpage.

4.Browser displays webpage.

Exam Point

- Web Server stores and serves websites.

HTML (HYPERTEXT MARKUP LANGUAGE)

HTML is the standard language used to create web pages.

Functions

- Creates webpage structure
- Displays text
- Displays images
- Creates hyperlinks

Exam Point

- HTML is not a programming language.
- It is a markup language.

INTERNET PROTOCOLS

A Protocol is a set of rules that allows devices to communicate.

Important Protocols

Protocol	Full Form	Function
TCP/IP	Transmission Control Protocol / Internet Protocol	Internet communication
HTTP	HyperText Transfer Protocol	Web communication
HTTPS	HyperText Transfer Protocol Secure	Secure web communication
FTP	File Transfer Protocol	Upload/Download files
SMTP	Simple Mail Transfer Protocol	Sending Email
POP3	Post Office Protocol v3	Receiving Email
IMAP	Internet Message Access Protocol	Managing Email
ICMP	Internet Control Message Protocol	Error reporting
TELNET	Telnet (protocol name, not a formal expanded acronym)	Remote login
SSH	Secure Shell	Secure remote login

PSTN (PUBLIC SWITCHED TELEPHONE NETWORK)

Provides traditional landline telephone services.

ISDN (INTEGRATED SERVICES DIGITAL NETWORK)

Provides voice and digital data communication using telephone lines.

DSL (DIGITAL SUBSCRIBER LINE)

Provides broadband Internet using telephone lines.

Exam Point

- DSL offers broadband services.

SSID (SERVICE SET IDENTIFIER)

SSID is the name of a Wi-Fi network.

Example

Home_Wifi

PunjabPolice_Wifi

NIC (NETWORK INTERFACE CARD)

Hardware component that enables a device to connect to a network.

Examples

- Ethernet Card
- Wi-Fi Adapter

Exam Point

- No NIC → No Network Connection

MAC ADDRESS (MEDIA ACCESS CONTROL ADDRESS)

A MAC address identifies a network interface on a local link. Addresses may be assigned by the manufacturer or set by software.

Characteristics

- Permanent
- Manufacturer assigned
- Designed for uniqueness, but duplication, spoofing and address randomization are possible

Example

00:1A:2B:3C:4D:5E

IMEI (INTERNATIONAL MOBILE EQUIPMENT IDENTITY)

A unique 15-digit number assigned to every mobile phone.

Uses

- Device Identification
- Mobile Tracking
- Blocking stolen devices

Exam Point

- IMEI contains 15 digits.

IMPORTANT FULL FORMS

- URL → Uniform Resource Locator
- WWW → World Wide Web
- HTML → HyperText Markup Language
- HTTP → HyperText Transfer Protocol
- FTP → File Transfer Protocol
- DNS → Domain Name System
- ISP → Internet Service Provider

- NIC → Network Interface Card
- MAC → Media Access Control
- IMEI → International Mobile Equipment Identity

MOST IMPORTANT FACTS FOR EXAMS

1.ARPANET was developed in 1969.

2.Internet services started in India on 15 August 1995.

3.TCP/IP is the backbone of the Internet.

4.DNS converts domain names into IP addresses.

5.URL is the address of a web resource.

6.WWW is different from the Internet.

7.HTML is used to create web pages.

8.IPv4 = 32-bit.

9.IPv6 = 128-bit.

10.FTP is used for file transfer.

11.HTTP is used for web communication.

12.SSID is the Wi-Fi network name.

13.MAC Address uniquely identifies a network device.

14.IMEI is a 15-digit mobile identification number.

15.NIC is required to connect a device to a network.

WEB BROWSER

A Web Browser is a software application used to access, retrieve, and display information available on the World Wide Web (WWW).

The information can be in the form of:

- Web Pages
- Images
- Videos
- Audio Files
- Documents
- Animations

A browser uses URLs (Uniform Resource Locators) and Hyperlinks to locate and display web resources.

Examples of Web Browsers

- Google Chrome
- Mozilla Firefox
- Microsoft Edge

- Safari
- Opera

What Does a Browser Do?

A browser acts as a client program.

User

Web Browser

Internet

Web Server

Requested Data

Browser Displays Web Page

Example

When you type:

www.facebook.com

The browser:

1.Sends a request to the server. 2.Receives webpage data.

3.Displays the webpage on your screen.

HISTORY OF WEB BROWSERS

1990 – WorldWideWeb (Nexus)

Created by: Tim Berners-Lee

- First web browser.
- Later renamed Nexus.

Exam Point

- First Web Browser = WorldWideWeb (Nexus)

1992 – Lynx

- Text-based browser.
- Could not display images or graphics.

Exam Point

- Lynx = Text Browser

1993 – NCSA Mosaic

- First popular graphical browser.
- Could display images and graphics.

Exam Point

- First widely popular graphical browser = Mosaic

1994 – Netscape Navigator

- Improved version inspired by Mosaic.
- Became very popular.

1995 – Internet Explorer

Introduced by: Microsoft

Exam Point

- First Microsoft Browser = Internet Explorer

1996 – Opera

- Publicly released.
- Lightweight browser.

2003 – Safari

Introduced by: Apple

Exam Point

- Default browser of Apple devices.

2004 – Firefox

Introduced by: Mozilla Foundation

2007 – Mobile Safari

- Mobile version of Safari.

2008 – Google Chrome

Introduced by: Google

Exam Point

- Most widely used browser worldwide.

2015 – Microsoft Edge

- Successor of Internet Explorer.

Exam Point

- Replaced Internet Explorer.

COMPONENTS OF A WEB BROWSER

1. User Interface (UI)

The visible area through which users interact with the browser.

Examples

- Address Bar
- Back Button
- Forward Button
- Refresh Button
- Bookmark Menu
- Tabs

Exam Point

- UI is the visible part of the browser.

2. Browser Engine

Acts as a bridge between:

- User Interface
- Rendering Engine

3. Rendering Engine

Function

Responsible for displaying web pages.

It interprets:

- HTML
- CSS
- XML
- Images

and displays them on the screen.

4. Networking Module

Handles communication over the Internet.

Uses protocols such as:

- HTTP
- HTTPS
- FTP

Responsibilities:

- Sending requests
- Receiving data
- Maintaining security
- Caching web pages

5. JavaScript Interpreter

Reads and executes JavaScript code.

Without it:

- Interactive websites would not work properly.

Examples

- Online forms
- Animations
- Dynamic content

6. UI Backend

Creates browser controls such as:

- Buttons
- Menus
- Checkboxes
- Dialog Boxes

7. Data Storage

Stores information locally.

Examples:

- Cookies
- Browser Cache
- Browsing History
- Saved Passwords

DOMAIN NAME

A Domain Name is a human-readable name used to identify a website on the Internet.

Instead of remembering IP addresses, users remember domain names.

Example: www.facebook.com

Domain Name:facebook.com

Why Domain Names Are Needed?

Difficult:157.240.22.35

Easy:facebook.com

Exam Point

- Domain names are easier to remember than IP addresses.

DOMAIN NAME STRUCTURE

Example:www.google.com

Part	Meaning
www	Subdomain
google	Second-level domain label; google.com is the domain name
.com	Top-Level Domain

TOP LEVEL DOMAINS (TLDs)

Generic Domains:

Domain	Meaning
.com	Commercial
.org	Organization
.edu	Educational Institution
.gov	Government
.mil	Military
.net	Network Organization
.biz	Business

Country Domains

Domain	Country
.in	India
.uk	United Kingdom
.us	United States
.au	Australia
.ca	Canada

Exam Point

- India's country domain = .in

WEB SURFING

The process of searching, browsing, and exploring information on websites is called Web Surfing.

Example

Searching:

- News
- Videos
- Educational Material
- Shopping Websites

Exam Point

- Web Surfing = Browsing Internet Resources

HOW DOES THE INTERNET WORK?

STEP-BY-STEP PROCESS

Step 1

User enters a website name.

Example:www.google.com

Step 2

Browser sends request.

Step 3

DNS converts domain name into IP address.

Example:

google.com

142.250.x.x

Step 4

Browser finds the web server using the IP address.

Step 5

Web server processes the request.

Step 6

Web server sends webpage files.

Step 7 Browser displays the webpage.

QUICK REVISION NOTES

First Browser :WorldWideWeb (Nexus)

First widely popular graphical browser: Mosaic

Most Popular Browser :Google Chrome

Microsoft Browser :Internet Explorer → Edge

Apple Browser :Safari

Browser Engines

- Chrome → Blink
- Firefox → Gecko
- Safari → WebKit
- IE → Trident

Domain Example :facebook.com

India's Domain :.in

Web Surfing :Searching and browsing websites.

MOST IMPORTANT FACTS FOR EXAMS

1.First web browser = WorldWideWeb (Nexus).

2.Created by Tim Berners-Lee in 1990.

3.First widely popular graphical browser = Mosaic (1993). 4.Internet Explorer was launched in 1995.

5.Google Chrome was launched in 2008.

6.Microsoft Edge was launched in 2015.

7.Chrome uses Blink rendering engine.

8.Firefox uses Gecko rendering engine.

9.Safari uses WebKit rendering engine.

10.Domain names identify websites.

11.DNS converts domain names into IP addresses.

12..gov = Government.

13..edu = Educational institutions.

14..com = Commercial organizations.

15..in = India.

16.Web Surfing means browsing Internet resources.